THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. At night the sea stays warmer than land because: 1

(a) sun shines **(b)** water cools slowly
(c) land is warm **(d)** sea has no air

2. Why do desert people wear loose white robes? 1

(a) absorb heat **(b)** look good
(c) reflect sun and let air circulate **(d)** keep out the dust

3. A person feels warmer in a black shirt than in a white shirt under the Sun because: 1

(a) black absorbs more radiation **(b)** white is hotter
(c) black reflects more **(d)** white absorbs more

4. A kite flies high on a hot day because: 1

(a) birds help **(b)** wind falls
(c) warm air rises and carries it **(d)** cold air rises

5. Why are ice boxes painted white or silver? 1

(a) look nice **(b)** absorb heat
(c) protect from cold **(d)** reflect heat

6. Which of the following needs a medium for heat transfer? 1

(a) sunshine **(b)** starlight **(c)** radiation **(d)** conduction

7. Smoke from a chimney goes up because: 1

(a) hot gases rise **(b)** fire is cold **(c)** wind is below **(d)** smoke is heavy

8. Why do houses in hot regions have white roofs? 1

(a) are cheaper **(b)** look dull **(c)** reflect sunlight **(d)** absorb heat

9. Which of the following is not heat transfer? 1

(a) conduction **(b)** evaporation **(c)** radiation **(d)** convection

10. Pot of water is heated from below, not from the top, because: 1

(a) convection spreads heat through the water
(b) it saves fuel
(c) the lid is hot
(d) top is cold

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we apply a shiny coat on roofs in hot places? 2

12. Why is the bottom of a pan blackened with soot but the inside is not? 2

13. Why do we feel the heat when we stand near a bonfire but not above it? 2

14. Why do two thin blankets keep a person warmer than one thick one? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an experiment to show that dark surfaces are better absorbers of heat. 3

16. Explain why water in a plastic bottle left in the sun gets warm. 3

17. Explain why ceramic or glass mugs keep tea warm longer than metal. 3

OR

How does the design of a solar cooker use the ideas of heat transfer? 3

18. Explain why a ceiling fan feels more comfortable in summer. 3

OR

Why do we blow on hot soup? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how sea breeze and land breeze form and why coastal places are pleasant. 5

OR

Explain what happens when you hold one end of a long metal rod and heat the other end, and what safety is required. 5

20. Discuss the importance of using different colours and fabrics for clothing and wall paint according to climate. 5

OR

Discuss how architects reduce heat in buildings in hot climates. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND